# PUSHPENDU KAR

***SFHEA(UK), FIETE(IN), CEng(UK), SMIEEE(US), SMACM(US)***


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**Present Address:** PMB 448, Sir Peter Mansfield Building, University of Nottingham Ningbo China, 199 Taikang East Road, Ningbo 315100, Zhejiang, China.

**Permanent Address:** Pushpendu Kar**,** C/o: Mihir Kumar Kar**,** Sonajhil, Sonarpur**,** 24–Parganas (South), Kolkata-700150, West Bengal, India.


## Career Objective

To be an asset to the organization through the knowledge, skill, and experience acquired during my professional sojourn.

## Research Interests

Ad Hoc Networks, Wireless Sensor Networks, Internet of Things, Content Centric Networking, Blockchain, BigData, and Quantum Communication.

## Academic Qualifications

- **Doctor of Philosophy (*PhD*),** Computer Science and Engineering
  Indian Institute of Technology Kharagpur, Kharagpur, West Bengal, India 721302
  Jan 2012 – Jan 2016.
  - Thesis: *Temporary Node Misbehavior in the Presence of Environmental Effects in Wireless Sensor Networks*
- **Master of Engineering (*M. E.*),** Computer Science and Engineering
  Jadavpur University, Kolkata, West Bengal, India 700032
  Jul 2008 – Jun 2010
  - Dissertation: *Implementation of a Jini-Based Service Framework in a Mobile Ad-Hoc Network*
- **Bachelor of Technology (*B. Tech*),** Computer Science and Engineering
  University of Kalyani, Kalyani, Nadia, West Bengal, India 741235
  Jun 2000 – May 2004
  - Dissertation: *Online Blood Pressure Measurement System*

## Professional Qualifications/Memberships

1. **Senior Fellow of Higher Education Academy (*SFHEA*), AdvanceHE**, UK, July 2023, Registration No.: PR269780
2. **Fellow,** The Institution of Electronics & Telecommunication Engineers **(*IETE*),** India, September 2023, Registration No.: F-504050
3. **Chartered Engineer (*CEng*), Engineering Council**, UK, April 2024, Registration No.: 709992
4. **Senior Member**, Association for Computing Machinery (***ACM***), USA, May 2026, Registration No.: 1355491
5. **Senior Member,** The Institute of Electrical and Electronics Engineers **(*IEEE*),** USA, July 2020, Registration No.: 93025879
6. **Sun Certified Java Programmer (*SCJP*) 5.0, Sun Microsystems**, USA, March 2009, Registration No.: SW409182

## Pedagogical Qualification

1. Postgraduate Certificate in Higher Education (*PGCHE*) [a master's level teaching qualification] by following the UK Professional Standards Framework (UKPSF) of AdvanceHE, UK, November 2023

## Professional Training

1. Completed eight weeks (21st May 2006 – 14th July 2006) Finishing School program from *NIT Durgapur, Durgapur, India,* conducted by MHRD in collaboration with *AICTE, Department of IT, NASSCOM, and IIT Roorkee, 2006*
2. Six months of training on J2EE from *"STG International (Regional Office)", Minto Park, Kolkata, India, 2005*
3. Three months of training on JAVA (Core & Advanced) from *"CMC Ltd.", Gariahat, Kolkata, India, 2005*

4. Three weeks (4th – 23rd January 2010) UGC sponsor refresher course on Wireless Ad-Hoc and Sensor Networks, Jadavpur University, Kolkata, India.
5. Six months of training on Hardware & Networking from *"Computer Foundation", Jadavpur, Kolkata, India, 2004.*

## Awards/Honours

1. Received Springer Nature's "Editor of Distinction Award 2026" for Editorial Contributions and Author Service to the Scientific Reports journal.
2. Received 'Faculty Star in Building International Network' award from the Faculty of Science and Engineering, University of Nottingham Ningbo China, 2025
3. Received 'Research Excellence' award from the Artificial Intelligence and Optimization (AIOP) research group at the University of Nottingham Ningbo China, 2025
4. Received 'Excellent Performance' awards for overall academic performance in the 2022-2023, 2023-2024, and 2025-2026 academic years at the University of Nottingham Ningbo China.
5. Received 'Spot Award' for teaching excellence and effective service to the University of Nottingham Ningbo China, 2021, 2025, and 2026
6. Received the 2020 IEEE Systems Journal (2020 I.F.: 4.463) Best Paper Award (this is one of the seven papers out of 793 papers [top 1%] that have received the award)
7. Received ERCIM Alain Bensoussan Fellowship award from the European Union to pursue a postdoc at *RISE SICS*, Sweden, 2018 (declined).
8. Received ERCIM Alain Bensoussan Fellowship award from the European Union to pursue a postdoc at the *Norwegian University of Science and Technology*, Norway, 2017.
9. Received SERB Overseas Postdoctoral Fellowship award from the Dept. of Science and Technology, Govt. of India to pursue a postdoc at *TU Delft*, Netherlands, 2017 (declined).
10. Received Erasmus Mundus Postdoctoral Fellowship award from the European Commission to pursue a postdoc at *Northumbria University*, Newcastle upon Tyne, United Kingdom, 2016 (declined).
11. Awarded a full financial institute travel grant to attend IEEE ICC 2015, London, United Kingdom.
12. Awarded a travel grant to attend XRCI Open 2015, Bangalore, India.
13. Awarded a travel grant to attend *The Fourth IDRBT Doctoral Colloquium*, Hyderabad, India, 2014.
14. Awarded the second prize in a state-level science quiz competition organized by Paschim Banga Vigyan Mancha, West Bengal, India, 1996.
15. Awarded the first prize from the Midnapore district in a state-level science exam competition organized by Paschim Banga Vigyan Mancha, West Bengal, India, 1995

## Distinguished Recognitions

1. Dr. Kar's project titled "AI-driven personalized learning paths" was featured in news coverage on Chinadaily.com. China Daily has the widest print circulation of any English-language newspaper in China.
2. Recognized as a High-Level Talent by the Ningbo Municipal Government, China, 2021

## Professional Experience (Total: 17+ years)

1. **Associate Professor** (Full-time), School of Computer Science
   **University of Nottingham, UK (China campus)**, 199 Taikang East Road, Ningbo 315100, China
   1st August 2025 – Present

2. **Assistant Professor** (Full-time), School of Computer Science
   **University of Nottingham, UK (China campus)**, 199 Taikang East Road, Ningbo 315100, China
   4th Match 2019 – 31st July 2025 (6 Yr 5 M)

3. **Research Fellow** (Full-time) [through ERCIM Fellowship], Department of ICT and Natural Sciences
   **Norwegian University of Science and Technology (NTNU)**, 6009 Ålesund, Norway
   01st February 2018 – 31st January 2019 (1 Yr)

4. **Post-Doctoral Research Fellow** (Full-time), Department of Electrical and Computer Engineering
   **National University of Singapore (NUS)**, Singapore 119077
   16th November 2016 – 31st January 2018 (1 Yr 2 M)

5. **Post-Doctoral Research Fellow** (Full-time), Energy Research Institute
   **Nanyang Technological University (NTU)**, Singapore 639798
   16th November 2015 – 15th November 2016 (1 Yr)

6. **Researcher** (Part-time), Department of Electronics and Electrical Communication Engineering
   **IIT Kharagpur**, Kharagpur, West Bengal, India 721302
   11th April 2014 - 28th August 2014 (4 M)

7. **Visiting Lecturer** (Part-time), AMIE Course
   **Institute of Engineers (Kharagpur Local Center)**, Kharagpur, West Bengal, India 721302
   15th October 2011 - 30th November 2013 (2 Yrs)

8. **Senior Project Assistant** (Full-time), Department of Computer Science and Engineering
   **IIT Kharagpur**, Kharagpur, West Bengal, India 721302
   14th October 2011 - 31st March 2013 (1 Yr 6 M)

9. **Associate System Engineer** (Full-time), Software Development
   IBM, Newtown, Rajarhat, Kolkata, India 700156
   17th June 2010 - 13th October 2011 (1 Yr 4 M)

10. **Lecturer** (Full-time), Department of Information Technology
    **Birbhum Institute of Engineering & Technology**, Birbhum, West Bengal, India 731101
    9th April 2006 - 31st July 2008 (2 Yrs 3 M)

11. **Lecturer** (Full-time), Department of Computer Science and Engineering
    **Siliguri Institute of Technology**, Sukna, Darjeeling, West Bengal, India 734009
    25th July 2005 - 8th April 2006 (9 M)

12. **Visiting Lecturer** (Part-time), Bachelor of Computer Applications
    **IT Center, North Bengal University**, Darjeeling, West Bengal, India 734013
    1st August 2005 - 8th April 2006 (9 M)

## Research Visit

1. MiMove Team, Inria Paris, France, 16-20 July 2018

## Courses Taught

- Programming and Algorithms, 5 Semesters
- Mobile Device Programming, 3 Semesters, Course coordinator
- Operating Systems, 3 Semesters, Course coordinator
- Data Communication and Networking, 2 Semesters, Course coordinator
- Object Oriented Programming using Java, 2 Semesters, Course coordinator
- Introduction to Computing, 4 Semesters, Course coordinator

## List of Publications

Google Scholar RG

### ❖ Refereed Journals:

1. X. Zhang, **P. Kar**, C. F. Kwong, D. Chieng, and A. G. Manzanares, "A Survey on the Security of 5G and Beyond Networks in the Post-Quantum Era", *Computer Networks (Elsevier)*, September 2026.

2. X. Zhang, **P. Kar**, C. F. Kwong, D. Chieng, and A. G. Manzanares, "A survey on Quantum Key Distribution: Foundations and future directions", *Computer Communications (Elsevier)*, August 2026, DOI: 10.1016/j.comcom.2026.108652

3. C. Yang, CF Kwong, D. Chieng, **P. Kar**, Q. Liu, and P. Fan, "Fog-assisted Proactive Scheduling for Platoon-based C-V2X Networks: System Architecture and Position-aware Resource Allocation", *Vehicular Communications (Elsevier)*, vol. 61, no. 101058, October 2026

4. W. Lyu, **P. Kar**, S. Welsen, and A. G. Manzanares, "COCO-QN: An Efficient Content-Aware Congestion Control Mechanism for Maintaining QoS in NDN", *IEEE Internet of Things Journal*, July 2026, DOI: 10.1109/JIOT.2026.3711147

5. Q. Mao, S. Xu, S. Chen, H. Zhou, **P. Kar**, and A. G. Bellotti, "High-Fidelity Text Refinement for ControlNet-Guided Latent Diffusion in Document Inpainting", *Expert Systems with Applications (Elsevier)*, vol. 332, Part A, no. 133491, July 2027, DOI: 10.1016/j.eswa.2026.133491

6. Z. Qiu, **P. Kar**, and U. Maulik, "A Genomic Data Analysis-Based Technique for Personalized and Precise Medicine", *ARRAY (Elsevier)*, vol. 30, pp. 100965, June 2026, DOI: 10.1016/j.array.2026.100965

7. J. C. Y. Lai, **P. Kar**, and U. Maulik, "When Posture Matters: Investigating Cuffless BP Estimation Errors During Postural Transitions", *Biomedical Signal Processing and Control (Elsevier)*, vol. 125, pp. 110685, June 2026, DOI: 10.1016/j.bspc.2026.110685

8. Z. Yang, Md. I. H. Gama, S. Abdullahi, M. Ali, D. Chieng, CF Kwong, **P. Kar**, and Z. Nie, "A dual-lock hierarchical mechanism empowered epifluidic patch for rapid glucose feedback in exercise-induced sweating", *Microchimica Acta*, vol. 193, no. 5, pp. 357, June 2026, DOI: 10.1007/s00604-026-08080-w

9. N. Chen, CF Kwong, D. Chieng, **P. Kar,** C. Yang, Q. Liu, and T. A. Tsiftsis, "Enhancing C-V2X with Blockchain and Zero-Knowledge Proofs for Improved Privacy, and Trustworthiness," *IEEE Transactions on Vehicular Technology*, March 2026, DOI: 10.1109/TVT.2026.3674840

10. X. Jin, D. Chieng, **P. Kar**, CF Kwong, Y. Li, and Y. Wang, "Large-Scale Continuous Monitoring of Greenhouse Gases with Adaptive LoRaWAN in CN470–510 MHz Band", *Sensors*, vol. 25, no. 17, pp. 5349, August 2025

11. F. Sun, P. Wang, Y. Zhang, and **P. Kar**, "βFSCM: An Enhanced Food Supply Chain Management System Using Hybrid Blockchain and Recommender Systems", *Blockchain: Research and Applications (Elsevier)*, vol. 6, no. 1, March 2025, DOI: 10.1016/j.bcra.2024.100245

12. Q. Mao, J. Li, H. Zhou, **P. Kar**, and A. G. Bellotti, "ALDII: Adaptive Learning-based Document Image Inpainting to Enhance the Handwritten Chinese Character Legibility of Human and Machine", *Neurocomputing (Elsevier)*, vol. 616, February 2025, DOI: 10.1016/j.neucom.2024.128897

13. **P. Kar** and Z. Wang, "Identifying COVID-19 Fake News on Social Networks Using Deep Learning: You will not know what happens next!", *WSEAS Transactions on Business and Economics*, vol. 21, pp. 1973-1987, October 2024, DOI: 10.37394/23207.2024.21.160

14. C. Yang, C. F. Kwong, D. Chieng, **P. Kar**, K. L. A. Yau, and Y. Chen, "Navigating the Road Ahead: A Comprehensive Survey of Radio Resource Allocation for Vehicle Platooning in C-V2X Communications", *IEEE Communications Surveys and Tutorials*, vol. 14, no. 8, June 2024, DOI: 10.1109/COMST.2024.3440033

15. **P. Kar**, L. Chen, W. Sheng, C. F. Kwong, and D. Chieng, "Advancing NDN Security: Efficient Identification of Cache Pollution Attacks through Rank Comparison", *Internet of Things (Elsevier)*, vol. 26, July 2024, DOI: 10.1016/j.iot.2024.101142

16. C. F. Kwong, C. Shi, Q. Liu, S. Yang, D. Chieng, and **P. Kar**, "Autonomous Handover Parameter Optimisation for 5G Cellular Networks using Deep Deterministic Policy Gradient", *Expert Systems With Applications (Elsevier)*, vol. 246, July 2024, DOI: 10.1016/j.eswa.2023.122871

17. L. Hu and **P. Kar**, "Design of a Blockchain-based Secure Health Monitoring System Using Decentralized Machine Learning Technique", *IEEE Communications Magazine,* vol. 62, no. 1, pp. 46 – 52, January 2024

18. **P. Kar** and S. Feng, "Intelligent Traffic Prediction by Combining Weather and Road Traffic Condition Information: a deep learning-based approach", *International Journal of Intelligent Transportation Systems Research (Springer)*, July 2023, DOI: 10.1007/s13177-023-00362-4.

19. X. Liang and **P. Kar**, "Robust and Secure File Transmission Through Video Streaming Using Steganography and Blockchain", *International Journal of Information and Computer Security (Inderscience)*, vol. 24, no. 3-4, pp. 258-283, September 2024, DOI: 10.1504/IJICS.2024.141610.

20. **P. Kar**, R. Chen, and Y. Qian. "An Efficient Producer Mobility Management Technique for Real-time Communication in NDN-based Remote Health Monitoring Systems", *Smart Health (Elsevier)*, vol. 26, December 2022, DOI: 10.1016/j.smhl.2022.100309.

21. **P. Kar**, K. Chen, J. Shi, "DMACN: A Dynamic Multi-Attribute Caching Mechanism for NDN-Based Remote Health Monitoring System", *IEEE Transactions on Computers*, vol. 72, no. 5, pp. 1301 – 1313, August 2022

22. **P. Kar**, A. Kumar, A. Shareef, K. T. Harn, and S. K. Panda, "An Intelligent Lighting Control System for Individual Visual Comfort and Energy Savings in Buildings", *Journal of Reliable Intelligent Environments (Springer)*, August 2022, DOI: 10.1007/s40860-022-00189-y.

23. D. Li and **P. Kar**, "*B-Spot*: Blockchain and Steganography based Robust and Secure Photo Transmission Mechanism", *Journal of Mobile Multimedia*, vol. 18, no. 6, pp. 1677–1708, July 2022.

24. **P. Kar**, Z. Xue, S. P. Ardakani, and C. F. Kwong, "Are Fake Images Bothering You on Social Network? Let's Detect Them Using Recurrent Neural Network", *IEEE Transactions on Computational Social Systems*, vol. 10, no. 2, pp. 783 – 794, March 2022

25. S. P. Ardakani, C. F. Kwong, **P. Kar**, Q. Liu, and L. LI, "CNN: a Cluster-based Named Data Routing for Vehicular Networks", *IEEE Access***,** vol. 9, pp. 159036-159047, November 2021.

26. Q. Liu, CF Kwong, W. Sun, S. Zhou, L. Li, and **P. Kar**, "Reinforcement Learning-Based Joint Self-Optimisation Method for the Fuzzy Logic Handover Algorithm in 5G HetNets", *Neural Computing and Applications (Springer)*, November 2021, DOI: 10.1007/s00521-021-06673-5

27. P. Zhao, **P. Kar**, and S. P. Ardakani, "ANTON: Activity Recognition Based Smart Home Control System", *SN Computer Science (Springer)***,** vol. 2, no. 428, pp. 1-12, August 2021, DOI: 10.1007/s42979-021-00824-0

28. L. Li, C. F. Kwong, Q. Liu, **P. Kar**, and S. P. Ardakani, "A Novel Cooperative Cache Policy for Wireless Networks", *Wireless Communications and Mobile Computing***,** vol. 2021, Article ID 5568935, 18 pages, August 2021. DoI:10.1155/2021/5568935

29. S. Misra, S. Goswami, C. Taneja, and **P. Kar**, "Heterogeneous Polydentate Mobile Chelating Node to Detect Breach in Surveillance Sensor Network", *Security and Privacy (wiley)***,** vol. 4, no. 6, pp. e175:1-22, June 2021.

30. **P. Kar**, S. Misra, A. Mandal, H. Wang, "SOS: NDN-Based Service-Oriented Game-Theoretic Efficient Security Scheme for IoT Networks", *IEEE Transactions on Network and Service Management***,** vol. 18, no. 3, pp. 3197-3208, Sept. 2021.

31. **P. Kar** and H. Wang, "EZPlugIn: Plug-n-Play Framework for a Heterogeneous IoT Infrastructure for Smart Home," *IEEE Internet of Things Magazine***,** vol. 4, no. 3, pp. 104-108, September 2021.

32. **P. Kar** and S. Misra, "On the Effects of Transfaulty Sensor Nodes in Stationary Wireless Sensor Network Systems," *IEEE Sensors Journal*, vol. 19, no. 13, pp. 5022 – 5029, March 2019

33. **P. Kar**, A. Shareef, A. Kumar, T. H. Koh, B. K. Mallikarjuna, and S. K. Panda, "ReViCEE: A Recommendation-based approach for personalized control of visual comfort & energy efficiency in buildings", *Building and Environment (Elsevier)*, vol. 152, pp. 135 – 144, April 2019

34. **P. Kar**, S. Misra, and M. S. Obaidat, "RILoD: Reduction of Information Loss in a WSN System in the Presence of Dumb Nodes," *IEEE Systems Journal*, vol. 13, no. 1, pp. 336 – 344, January 2018. (**Received 2020 IEEE Systems Journal Best Paper Award**).

35. **P. Kar**, A. Roy, S. Misra, and M. S. Obaidat, "On the Effects of Communication Range Shrinkage of Sensor Nodes in Mobile Wireless Sensor Networks Due to Adverse Environmental Conditions," *IEEE Systems Journal*, vol. 12, no. 3, pp. 2048-2055, September 2018.

36. A. Kumar, **P. Kar**, R. Warrier, A. Kajale, and S. K. Panda, "Implementation of Smart LED Lighting and Efficient Data Management System for Smart Buildings," *Energy Procedia (Elsevier)*, vol. 143, pp. 173-178, December 2017.

37. A. Roy, S. Misra, **P. Kar**, and A. Mondal, "Topology Control for Self-Adaptation in Wireless Sensor Networks with Temporary Connection Impairment," *ACM Transactions on Autonomous and Adaptive Systems*, vol. 11, no. 4, pp. 1-34, February 2017.

38. **P. Kar** and S. Misra, "Reliable and Efficient Data Acquisition in Wireless Sensor Network in the Presence of Transfaulty Nodes," *IEEE Transactions on Network and Service Management*, vol. 13, no. 1, pp. 99-112, January 2016.

39. **P. Kar**, A. Roy, and S. Misra, "Connectivity Re-establishment in Self-organizing Sensor Networks with Dumb Nodes," *ACM Transactions on Autonomous and Adaptive Systems*, vol. 10, no. 4, pp. 1-30, February 2016.

40. A. Roy, **P. Kar**, S. Misra, and M. S. Obaidat, "D3: Distributed Approach for the Detection of Dumb Nodes in Wireless Sensor Networks," *International Journal of Communication Systems (Wiley),* vol. 30, no. 1, pp. e2913, January 2015

41. **P. Kar** and S. Misra, "Detouring Dynamic Routing Holes in Stationary Wireless Sensor Networks in the Presence of Temporarily Misbehaving Nodes," *International Journal of Communication Systems (Wiley)*, vol. 30, no. 4, pp. e3009, June 2015

42. S. Misra, **P. Kar**, A. Roy, and M. S. Obaidat, "Existence of Dumb Nodes in Stationary Wireless Sensor Network," *Journal of Systems and Software (Elsevier),* vol. 91, pp. 135–146, May 2014.

## ❖ International Conferences / Workshops

1. X. Zhang, **P. Kar**, CF Kwong, D. Chieng, A. G. Manzanares, U. Maulik, and S. Bandyopadhyay, "MDI-QKD-Assisted Quantum-Safe User Plane for 5G and Beyond Networks", *in Proceedings of the International Symposium on Quantum Intelligence (SEQUENCE 2026)*, Ostrava, Czech Republic, 22 - 23 October 2026 (Accepted on 8th October 2026)

2. F. Lu and **P. Kar**, "A Blockchain-Based Consent Management System with Hybrid Transformer for Privacy-Preserving Healthcare AI", *in Proceedings of the IEEE Blockchain Conference 2026*, Montbéliard, France, 30th November - 4th December 2026 (Accepted on 16th September 2026)

3. H. Geng, **P. Kar**, and E. Zahran, "Class-Level Wellbeing Profiling and Fuzzy Teaching-Plan Generation from Social-Media Text: A Non-Clinical Prototype", *the IEEE International Conference on Teaching, Assessment, and Learning for Engineers (TALE 2026)*, 2 - 4 December 2026, Pattaya, Thailand

4. F. B. Tesema, **P. Kar**, D. Towey, T. Cui, and M. Solomon, "From Information Retrieval to AI-Augmented Learning: A Qualitative Study of Computer Science Students' Experiences with Generative AI", *the IEEE International Conference on Teaching, Assessment, and Learning for Engineers (TALE 2026)*, 2 - 4 December 2026, Pattaya, Thailand

5. X. Zhang, **P. Kar**, CF Kwong, D. Chieng, T. Tsiftsis, and A. G. Manzanares, "Securing 5G User Plane Using Post-Quantum DTLS Protocol", *in Proceedings of the 31st IEEE Symposium on Computers and Communications (ISCC 2026)*, Vilamoura, Algarve, Portugal, 23 - 26 June 2026 (Accepted on 9th April 2026)

6. **P. Kar** and Y. Chen, "BDWeb: Blockchain-based Distributed Web Hosting System using IPFS", *in Proceedings of the 5th International Conference on Cyber Security, Artificial Intelligence and Digital Economy (CSAIDE 2026)*, Salamanca, Spain, 13 – 15 March 2025

7. J. Lai, **P. Kar**, "A Multistage Signal Quality Framework for Blood Pressure Monitoring Using Photoplethysmography", *in Proceedings of the 25th IEEE International Conference on Bio-Informatics and Bio-Engineering (BIBE 2025)*, pp. 429-435, Athens, Greece, 6 - 8 November 2025

8. J. Lai, **P. Kar**, "Synchronization of Wearable Sensor Data for Vital Sign Monitoring", *in Proceedings of the 25th IEEE International Conference on Bio-Informatics and Bio-Engineering (BIBE 2025)*, pp. 99-105, Athens, Greece, 6 - 8 November 2025

9. W. Lyu, **P. Kar**, S. Welsen, and A. G. Manzanares, "ProLoC: An Efficient Probability-Based Forwarding Mechanism Considering Load Factor for Congestion Control in Named Data Networking", in proceedings of the *IEEE International Conference on Internet of Things (iThings 2025)*, pp. 44-51, Zhengzhou, Henan, China, October 30 - November 02, 2025

10. N. Chen, CF Kwong, D. Chieng, **P. Kar**, Z. Chu, and P. Fan, "Efficient and Secure Data Sharing in Scalable C-V2X with Dynamic Sharding Blockchain and Zero-Knowledge Proofs", *in proceedings of the IEEE International Conference on Communications (ICC 2025)*, pp. 2918-2924, Montreal, Canada, 8 - 12 June 2025

11. Y. Zhu and **P. Kar**, "An Accurate Salary Estimation Scheme by Using BigData Technique", *in proceedings of the 11th IEEE/ACM International Conference on Big Data Computing, Applications, and Technologies (BDCAT 2024)*, pp. 1 - 3, Sharjah, United Arab Emirates, 16 - 19 December 2024

12. N. Chen, C. F. Kwong, D. Chieng, and **P. Kar**, "Efficient Data Processing in C-V2X with Dynamic Sharding Blockchain and Zero-Knowledge Proofs", *in proceedings of International Conference on Internet of Things (ICIoT 2024)*, pp. 41-46, Ningbo, China, 11-12 October 2024

13. **P. Kar** and P. Ingkasit, "Development of a User-Friendly and Efficient Control System for Smart Home", *in proceedings of IEEE International Conference on Internet of Things (iThings 2024)*, pp. 288-293, Copenhagen, Denmark, 19 – 22 August 2024

14. Z. Bi and **P. Kar**, "BUMS: A Novel Balanced Multi-model Machine Learning System for Real-Time Blood Glucose Prediction and Abnormal Glucose Events Detection", *in proceedings of IEEE International Conference on E-health Networking, Application & Services (IEEE Healthcom 2023),* pp. 13-18, Chongqing, China, 15–17 December 2023

15. I. Cho, D. Towey, and **P. Kar**, "Using Obfuscators to Test Compilers: A Metamorphic Experience", *in proceedings of the 15th IEEE International Workshop on Software Test Automation in conjunction with IEEE COMPSAC 2023*, pp. 1786-1791, Torino, Italy, 26-30 Jun 2023

16. R. Arathy, B. Bhabani, M. Malakar, J. Mahapatro, and **P Kar**, "Jamming Attack Classification in Wireless Networks using Machine Learning", *in Proceedings of the 4th International Conference on Communication and Intelligent Systems (ICCIS 2022)*, pp. 477–491, New Delhi, India, 19-20 Dec 2022.

17. Z. Wang, Y. Guo, T. Xia, B. Ye, and **P. Kar**, "Detection of COVID-19 Through Thermal and Voice Sensing Using Smartphone", *in proceedings of the 22nd IEEE/ACIS International Conference on Computer and Information Science (ICIS 2022)*, pp. 2-7, Zhuhai, China, 26-28 Jun 2022

18. L. Hu, Y. Huang, and **P. Kar**, "An Integrated Single Device Framework for Combined Face Recognition and Covid-19 Detection Using Thermal Infrared Imagery", *in proceedings of the 22nd IEEE/ACIS International Conference on Computer and Information Science (ICIS 2022)*, pp. 8-13, Zhuhai, China, 26-28 Jun, 2022

19. S. Datta, M. Roy, **P. Kar**, "Application of IoT in Smart Epidemic Management in context of Covid-19", *in proceedings of the 22nd IEEE International Conference on High Performance Switching and Routing (HPSR 2021)*, pp. 1-6, Paris, France, Jun 2021.

20. J. Shi, W. Sheng, **P. Kar**, M. Roy, S. Datta, "A Novel Framework for Predicting the Spread of COVID-19 by Contact Tracing through Smartphone", *in proceedings of the 17th IEEE International Wireless Communications and Mobile Computing Conference (IWCMC 2021)*, pp. 570-575, Harbin, China, Jun 2021.

21. R. Chen, Y. Zhang, Y. Fei, and **P. Kar**, "WLEACH-CK: Weighted K-Means Based LEACH-C Algorithm for Cluster Head Selection", *in proceedings of the 17th IEEE International Conference on the Design of Reliable Communication Networks (DRCN 2021)*, pp. 1-6, Milan, Italy, April 2021.

22. **P. Kar**, Y. Dong, Y. Li, X. Ma, X. Ding, Y. Wang, "NDN-Based Plug-n-Play and Secure Remote Health Monitoring System", *in proceedings of the IEEE International Conference on Communication (ICC 2021)*, pp. 1-6, Montreal, Canada, Jun 2021.

23. Z. Song and **P. Kar**, "Name-Signature Look Up System: A Security Enhancement to Named Data Networking", *in proceedings of the 19th IEEE International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom 2020)*, pp. 1444-1448, Guangzhou, China, Dec 2020.

24. S. Wu, Y. Yuan, and **P. Kar**, "Lightweight Verification and Fine-grained Access Control in Named Data Networking Based on Schnorr Signature and Hash Functions", i*n proceedings of the 20th IEEE International Conference on Communication Technology (ICCT 2020)*, pp. 1561-1566, Nanning, China, Oct 2020.

25. A. Kumar, A. Shareef, K. T. Harn, **P. Kar**, and S. K. Panda, "A Complete Hardware Setup for Smart Lighting System", *in Proceedings of the IEEE International Conference on Sustainable Energy Technologies and Systems (ICSETS 2019)*, pp. 297-300, Bhubaneswar, India, Feb 2019

26. **P. Kar**, S. Misra, A. Mandal, and H. Wang, "SecureIoT: Hop-Count Based Service-Oriented Efficient Security Solution for IoT", *in Proceedings of the ACM International Workshop on Future Industrial Communication Networks (FICN 2018) in conjunction with ACM MobiCom 2018*, pp. 15-20, New Delhi, India, Nov 2018.

27. **P. Kar** and B. Dappuri, "Site Survey and Radio Frequency Planning for the Deployment of Next Generation WLANs", *in Proceedings of the IEEE International Conference on Wireless Communications, Signal Processing and Networking (WiSPNET 2018)*, pp. 1-4, Chennai, India, 2018.

28. **P. Kar**, S. Das, and A. Kumar, "PreCoRe: Predictive Connectivity Reestablishment in Wireless Sensor Network in the Presence of Dumb Nodes," *in Proceedings of the IEEE Region 10 Conference (TENCON 2017)*, pp. 403-408, Penang, Malaysia, November 2017.

29. A. Kumar, A. Kajale, **P. Kar**, R. Warier, A. Shareef, S. K. Panda, "Location-aware Smart Lighting System for Individual Visual Comfort in Buildings," *in Proceedings of the 6th IEEE Global Conference on Consumer Electronics (GCCE 2017)*, pp. 1-2, Nagoya, Japan, October 2017.

30. A. Kumar, A. Kajale, **P. Kar**, R. Warrier, and S. K. Panda, "Implementation and Integration of a Smart App in a Smart Building for Personal Visual Comfort," *in Proceedings of the 12th IEEE International Conference on Power Electronics and Drive Systems (PEDS 2017)*, pp. 1161-1166, Honolulu, Hawaii, USA, December 2017.

31. S. Das, **P. Kar**, and S. Barman, "Analysis of IEEE 802.11 WLAN Frame Aggregation Under Different Network Conditions," *in Proceedings of the IEEE International Conference on Wireless Communications, Signal Processing and Networking (WiSPNET 2017)*, pp. 1240-1245, Chennai, India, March 2017.

32. **P. Kar**, S. Barman, and S. Das, "ISCP: Inter-state changing problem in Wireless Sensor Network with dumb nodes," *in Proceedings of the International Conference on Computing and Communication Systems (I3CS 2016)*, Shillong, India, November 2016, DoI:10.1007/978-981-10-6890-4_2.

33. S. Das, **P. Kar**, and D. K. Jana, "SDH: Self-detection and Healing mechanism for dumb nodes in Wireless Sensor Network", *in Proceedings of the IEEE Region 10 Conference (TENCON 2016)*, pp. 2792-2795, Singapore, November 2016.

34. **P. Kar**, R. S. Reddy, P. Gupta, J. Dauwels, and A. Ukil, "FLoGPN: A Reputation-Based Scheme for Fault Localization in Gas Pipeline Network," *in Proceedings of the 42nd Annual Conference of IEEE Industrial Electronics Society (IECON 2016)*, pp. 4651-4656, Florence, Italy, October 2016 (**Received best presentation award**).

35. **P. Kar**, A. Roy, S. Misra, and M. S. Obaidat, "Energy-Efficient Connectivity Re-establishment in WSN in the Presence of Dumb Nodes," *in Proceedings of the IEEE ICC 2015 — 4th IEEE International Workshop on Smart Communication Protocols and Algorithms (SCPA)*, pp. 1485 – 1490, London, UK, June 2015.

36. A. Roy, **P. Kar**, and S. Misra, "Detection of Dumb Nodes in a Stationary Wireless Sensor Network," *in Proceedings of the 11th IEEE India Conference (INDICON 2014)*, pp. 1–6, Pune, Maharashtra, India, December, 2014.

37. M. Roy, **P. Kar**, and N. Mukherjee, "A Jini Based Implementation for Best Leader Node Selection in MANETs," *in Proceedings of the International Conference on Computer Networks & Communications (NetCom 2012),* pp. 15 - 22, Chennai, India, December 2012.

38. M. Roy, **P. Kar**, and N. Mukherjee, "An Enhanced Service Framework in MANETs for Application in Emergency Services," *in Proceedings of the IEEE International Conference on Intelligent Network and Computing (ICINC 2010),* no.1, pp.191-198, Kuala Lumpur, Malaysia, November 2010.

39. M. Roy, **P. Kar**, and N. Mukherjee, "Determining JINI Leasing Time Limits Using the Random Waypoint Mobility Model in Mobile Ad Hoc Networks," *in Proceedings of the 10th IEEE International Conference on Networking and Information Technology (ICNIT 2010),* pp. 64-69, Manila, Philippines, June 2010.

## ❖ Books

### ➢ Authored

1. **P. Kar,** M. Roy, and S. Dutta, "Recommender Systems: Algorithms and their Applications", Springer, June 2024, ISBN: 978-981-97-0537-5 [Link]

### ➢ Edited

1. M. Roy, **P. Kar**, and S. Dutta, "Recommender Systems: A Multi-Disciplinary Approach", *CRC Press*, June 2023, ISBN: 9781003319122 [Link]
2. M. Roy, **P. Kar**, and S. Dutta, "Interoperability in IoT for Smart Systems", *CRC Press*, December 2020, ISBN: 9780367519865 [Link]
3. G. G. Md. Nawaz Ali, Md. N. A. Rahim, and **P. Kar**, "Vehicular Ad-Hoc Networks: Applications and Technology", Nova Science Publishers, September 2020, ISBN: 978-1-53618-038-1 [Link]

### ➢ Proceedings

1. **P. Kar**, J. Li, Y. Qiu, "Proceedings of the 2023 International Conference on Image, Algorithms and Artificial Intelligence (ICIAAI 2023)", Advances in Computer Science Research series, Atlantis Press, ISBN: 978-9464632996 [Link]

## ❖ Book Chapters

1. Z. Yang, D. Chieng, CF Kwong, **P. Kar**, and Z. Nie, " Colorimetric sensing platforms for noninvasive glucose monitoring", *in Emerging Diagnostic Glucose Sensing Approaches*, Elsevier, January 2026, DOI: 10.1016/B978-0-443-36348-1.00008-6
2. S. Singh, M. Roy, S. Datta, and **P. Kar**, "Application of Swarm Intelligence in Recommender Systems", *in Recommender Systems: A Multi-Disciplinary Approach*, CRC Press, June 2023, DOI: 10.1201/9781003319122
3. A. Saha, S. Dey, M. Roy, S. Datta, **P. Kar**, "Recommendation Systems for Choosing Online Learning Resources: A Hands-On Approach", *in Recommender Systems: A Multi-Disciplinary Approach*, CRC Press, June 2023, DOI: 10.1201/9781003319122
4. **P. Kar**, "Optimal Hole Coverage in the Presence of Dumb Node in Wireless Sensor Networks", in *Vehicular Ad-Hoc Networks: Applications and Technology*, Nova Science Publishers, September 2020
5. **P. Kar** and A. Das, "Artificial Neural Networks and Learning Techniques", in P. Samui (Eds.), *Advanced Computational Techniques for Simulation-Based Engineering*, *IGI Global, USA*, DOI: 10.4018/978-1-4666-9479-8.ch009, March 2016.

## Patents

1. X. Zhang, **P. Kar**, C. F. Kwong, D. Chieng, and A. G. Manzanares, "Post-quantum secure transmission methods, systems, devices and media for 5G user plane", PCT patent filed on 15th April 2026 (Patent no: PCT/CN2026/097720)

2. Q. Mao, A. G. Bellotti, **P. Kar**, "A method and system for document image repair", Chinese patent filed on 31st October 2025 (Patent no: 202511575672.X)

3. Q. Mao, A. G. Bellotti, **P. Kar**, "An enhanced method and system for text recognition in document images", Chinese patent filed on 4th November 2025 (Patent no: 202511592626.0)

4. X. Zhang, **P. Kar**, C. F. Kwong, D. Chieng, and A. G. Manzanares, "Post-quantum secure transmission methods, systems, devices and media for 5G user plane", Chinese patent **granted** on 11th September 2026 (Patent no: 202511629269.0)

5. Q. Mao, A. G. Bellotti, **P. Kar**, "A text image repair method, device and storage medium", Chinese patent **granted** on 27th February 2025 (Patent no: 202410370947.5)

6. S. Misra, S. Goswami, **P. Kar**, and A. Roy, "PKI Enabled Time Stamped Digital Signing System Involving Certification Authority Issued Digital Certificate Cryptographic Token With Real-Time Revocation Verification", Indian patent **granted** on 24th June 2024 (Patent no: 537779).

7. S. Misra, A. Roy, **P. Kar**, and S. Goswami, "An Adverse Environmental Effect Resistant Seamless Wireless Sensor Network System", Indian patent **granted** on 12th December 2023 (Patent no: 480946).

8. S. Misra, **P. Kar**, A. Roy, and S. Goswami, "An Advanced Wireless Sensor Network System and Method for Accurate Information Gathering from a Radiation Affected Area", Indian patent **granted** on 3rd August 2023 (Patent no: 442698).

9. **P. Kar** and L. Hu, "A Health Monitoring System", Chinese patent filed on 23rd September 2022 (Patent no: 202211168088.9)

10. A. Kumar, A. Shareef, **P. Kar**, B. K. Mallikarjuna, T. H. Koh, and S. K. Panda, "An Intelligent Lighting System for Smart Buildings Using Multi-Agent-Enabled Wireless Sensor-Actuator Network", Singapore patent filed on 1st June 2018 (Patent no: 10201804681P)

## Research Grants Received

1. **Project title:** AI-driven personalized learning paths
   **Sponsoring agency:** Ministry of Education China, Sail New Technology (Beijing) Co. Ltd., and Qi An Xin Technology Group Co. Ltd. (China University Industry-University-Research Innovation Fund for the AI+Cybersecurity Governance Technology Special Project)
   **Role:** Principal Investigator (PI)
   **Budget:** ¥150,000
   **Duration:** 2026-2027 (Ongoing)

2. **Project title:** Sustainable Waste Management System by Integrating Internet of Things (IoT) Sensors with Blockchain Technology
   **Sponsoring agency:** China International Talent Exchange Foundation
   **Role:** Principal Investigator (PI)
   **Budget:** ¥140,000
   **Duration:** 2025-2026 (Completed)

3. **Project title:** PCD-HWR: Passive, Contactless, Deviceless AI-driven Hybrid WiFi-mmWave Radar System for Patient Rehabilitation Therapies
   **Sponsoring agency:** University of Nottingham Ningbo China (Interdisciplinary Research Engagement Fund)
   **Role:** Co-Investigator (CoI)
   **Budget:** ¥60,000
   **Duration:** 2025-2026 (Ongoing)

4. **Project title:** Research on Adaptive Radio Resource Allocation for Vehicle Platooning in 5G Networks: A DDPG and NOMA-Based Approach
   **Sponsoring agency:** Ningbo Natural Science Foundation, China
   **Role:** Co-Investigator (CoI)
   **Budget:** ¥48,000
   **Duration:** 2025-2027 (Ongoing)

5. **Project title:** Research on credit risk assessment and decision support application based on Ningbo financial intelligence
   **Sponsoring agency:** Ningbo Municipal Government, China (Yongjiang Talent Scheme)
   **Role:** Co-Investigator (CoI)
   **Budget:** ¥1,000,000
   **Duration:** 2022-2026 (Ongoing)

6. **Project title:** DOMINANT: Development of an Efficient Plug-n-Play and Real-Time Remote Health Monitoring System
   **Sponsoring agency:** Ningbo Science &Technology Bureau, China (Commonweal Research Programme)
   **Role:** Principal Investigator (PI)
   **Budget:** ¥200,000
   **Duration:** 2020-2022 (Completed)

7. **Project title:** Implementation of Named Data Networking Based Plug-n-Play and Secure Framework for Heterogeneous IoT Network
   **Sponsoring agency:** University of Nottingham Ningbo China (Faculty Inspiration Grant)
   **Role:** Principal Investigator (PI)
   **Budget:** ¥30,000
   **Duration:** 2019-2021 (Completed)

8. **Project title:** EZPlugIn: Universal Plug-n-Play Framework in a Heterogeneous IoT Infrastructure
   **Sponsoring agency:** European Union (ERCIM Fellowship Programme)
   **Role:** Principal Investigator (PI)
   **Budget:** €46,000
   **Duration:** 2018-2019 (Completed)

## Research Projects Involved In:

1. **Title:** Intelligent Information Management System in Smart Buildings Using Multi-Agent-Enabled Wireless Sensor-Actuator Network
   **Sponsoring Agency:** Building & Construction Authority, Singapore
   **Description:** In this project, we aim at innovations in the intelligent information system for building Operation and management. This is performed using novel technologies of Wireless Sensor Actuator Network (WSAN) and Multi-Agent Systems (MAS).
   **Technology Used:** MATLAB, JAVA, Python, Processing Code, PCB Design, Sensor Technology
   **Institution:** National University of Singapore (NUS), Singapore
   **Duration:** 2016 - 2018
   **Role:** Postdoctoral Research Fellow (Contributions: System design, simulation, implementation, and deployment)

2. **Title:** Advanced Multi-Sensor Anomaly Monitoring and Analytics for Gas Pipeline
   **Sponsoring Agency:** Energy Management Authority, Singapore
   **Description:** In this project, my focus was to explore possible detection techniques for identifying, analyzing, and then locating anomalies (such as leakage, water ingress, deformation, etc.) of underground natural gas pipelines, as well as perform advanced data analytics based on predictive maintenance for the possible future failure-prone zones.
   **Technology Used:** MATLAB, COMSOL, OpenFOAM
   **Institution:** Nanyang Technological University (NTU), Singapore.
   **Duration:** 2015 - 2016
   **Role:** Postdoctoral Research Fellow (Contributions: System design, simulation, implementation, and deployment)

3. **Title:** Checking Border Violation by Civilians Through Technological Solution: Phase I
   **Sponsoring Agency:** Border Security Force (BSF), Govt. of India
   **Description:** This project includes a survey of related works, proposes new solutions for automated Checking border violations, documentation, and writing a proposal.
   **Technology Used:** Wireless Sensor Network (WSN) and video camera
   **Institution:** Indian Institute of Technology Kharagpur (IIT Kharagpur), West Bengal.
   **Duration:** April - August, 2014
   **Role:** Researcher (Contributions: Survey related works, proposing solutions, documentation, and report writing)

4. **Title:** Towards Robust, Efficient, and Secure Data Acquisition in Underwater Sensor Networks (TRE)
   **Sponsoring Agency:** Department of Information Technology(DIT), Govt. of India
   **Description:** Design and implementation of network protocols toward the development of an Underwater acoustic network simulator
   **Technology Used:** Network Simulator Version 3 (NS-3), MATLAB
   **Institution:** Indian Institute of Technology Kharagpur (IIT Kharagpur), West Bengal, India
   **Duration:** 2011 - 2013
   **Role:** Senior Project Assistant (Contributions: Simulator Development using NS-3)

## Other Projects Involved In:

1. **Title:** Walmart Purchase Order Transformation
   **Client Name:** Walmart Americans.
   **Description:** This project developed an online system to create and edit purchase agreements and Purchase orders for suppliers by adding item information with supplier information and send them to the supplier.
   **Technology Used:** Java, J2EE, JSF, and DB2.
   **Organization:** IBM India Pvt. Ltd. (Global Business Service)
   **Duration:** 2010 - 2011
   **Role:** Associate System Engineer (Contributions: System design and development)

2. **Title:** Event Management System
   **Description:** Developed an online application for managing all the activities of an educational institute. This project has three major modules such as Administrator, Speaker, and Participant.
   **Technology Used:** JSP using Struts framework & DB2 database
   **Organization:** IBM India Pvt. Ltd. (Global Business Service)
   **Duration:** June – August, 2010
   **Role:** Associate System Engineer (Contributions: System design and development)

3. **Title:** Data Tracking System (version-1.0)
   **Description:** This project helps to computerize and integrate the flow of operations of the Crime Record Branch and Central Fingerprint Branch in the National Crime Records Bureau
   **Technology Used:** VISUAL BASIC 6.0, MS-Access
   **Organization:** National Crime Records Bureau (NCRB), Ministry of Home Affairs, New Delhi, India.
   **Duration:** March – May, 2004
   **Role:** Internship student (Contributions: System design, writing programs using Visual Basic 6.0)

## Membership

1. Institute of Electrical and Electronics Engineers (IEEE), Senior Member, ID: 93025879
2. Association for Computing Machinery (ACM), Senior Member, ID: 1355491
3. Institution of Engineering and Technology (IET), Member, ID: 1101015556
4. International Association of Computer Science and Information Technology (IACSIT), Lifetime Member, ID: 80339962

## Activities

### ❖ Professional Institution Leadership

1. International Professional Registration Advisor (IPRA), The Institution of Engineering and Technology (IET), UK, 2024 - Present.
2. Professional Registration Interviewer (PRI) for CEng/IEng, The Institution of Engineering and Technology (IET), UK, 2024 - Present.

### ❖ Research Laboratory Development

1. ***Co-Developer***, Next Generation Internet of Things (NGIoE) Laboratory, Faculty of Science and Engineering, University of Nottingham Ningbo, China (Directors: Dr. Chiew-Foong Kwong and Co-Director: Dr. David Chieng)

### ❖ Editorial Activities

1. ***Guest Editor***, Industrial uses and benefits of blockchain applications, Scientific Reports (Nature portfolio, Impact factor: 4.3), 2025 - 2026
2. ***Editorial Board Member***, International Journal of Applied Pattern Recognition (Impact factor: 1.1), InderScience, 2025 - Present
3. ***Editorial Board Member***, Scientific Reports (Impact factor: 3.8), Springer Nature, 2025 - Present
4. ***Guest Editor***, PLOS ONE, 2024 – Present
5. ***Editorial Board Member,*** Computing and Artificial Intelligence, 2024 - 2026
6. ***Guest Editor***, Internet of Medical Things (IoMT): Current Challenges to Future Solutions with Innovative Techniques and Tools for Smart Healthcare Applications, Frontiers in Communications and Networks (Impact factor: 2.21), 2023 - 2024.
7. ***Editorial Board Member,*** International Journal of Engineering Business Management, Sage Publisher (Impact factor: 3.9), 2022 - Present
8. ***Editor***, American Journal of Networks and Communications, Science Publishing Group, 2022 – Present
9. ***Review Editor***, Frontiers in the Internet of Things - IoT Services and Applications, 2022 – Present

10. ***Academic Editor,*** Journal of Healthcare Engineering, Hindawi (Impact factor: 2.68), 2021 - 2023

## ❖ Conference Organization:

1. Organizing the 2nd IEEE International Conference on Data Science and Edge Computing (ICDSEC 2026), 4 - 6 December 2026, Ningbo, China, Role: General Chair
2. Organized the 4th IEEE International Conference on Internet of Things and Cloud Computing Technology (IoTCCT 2026), 11 - 13 September 2026, Ningbo, China, Role: General Chair
3. Organized IEEE International Conference on Data Science and Edge Computing (ICDSEC 2025), 6 - 8 December 2025, Changsha, China, Role: General Chair
4. Organized the International Conference on Internet of Things (ICIoT 2024), 24 - 25 October 2024, Ningbo, China, Role: Technical Program Committee Chair

## ❖ Distinguished Talks:

1. ***Keynote Speech***: "Intelligent Traffic Prediction by Combining Weather and Road Traffic Condition Information: A Deep Learning-Based Approach", International Conference on Artificial Intelligence, New Materials and New Energy (AINMNE 2026), Xiangtan, China, 24th October 2026
2. ***Special Guest Speech***: "Learner-Based Teaching Plan Through AI-Assisted Analysis of Students' Mental Health Using Social Media Data", Asia-Pacific Artificial Intelligence Education Conference (AAIEC 2026), Fuzhou, China, 23rd May 2026
3. ***Plenary Speech***: "Blockchain-based Distributed Web Management", 10th International Conference on Decision Science and Management (ICDSM 2026), Hong Kong, China, 25th April 2026
4. ***Keynote Speech***: "Named Data Networking: Future of the Internet", 3rd International Conference on Artificial Intelligence, Systems and Network Security (AISNS 2025), Xiangtan, China, 27th December 2025
5. ***Keynote Speech***: "Named Data Networking: Future of the Internet", International Conference on Data Science and Edge Computing (ICDSEC 2025), Changsha, China, 7th December 2025
6. ***Invited Talk***: "Identifying COVID-19 Fake News on Social Networks Using Deep Learning: You will not know what happens next!", the 8th International Conference on Mathematical Models & Computational Techniques in Science & Engineering (MMCTSE 2024), Athens, Greece, 02nd June 2024
7. ***Invited Talk***: "Smart Home Control Through Activity Recognition", Techno College of Engineering Agartala, Tripura, India, 22nd May 2024
8. ***Keynote Speech***: "Enhancing NDN Security: Name Signature Lookup System", 13th International Conference of Information and Communication Technology (ICTech 2024), Xiamen, China, 13th April 2024
9. ***Keynote Speech***: "Activity Recognition Based Smart Home Control System", International Conference on Image, Algorithms and Artificial Intelligence (ICIAAI 2023), Singapore, 11th August 2023
10. ***Keynote Speech***: "Activity Recognition Based Smart Home Control System", International Conference on Mechatronics and Automation Technology (ICMAT 2022), Wuhan, China, 29th October 2022
11. ***Keynote Speech***: "Name Signature Lookup System for Enhanced Security of NDN-based IoT Networks", 2nd International Conference on Computer, Remote Sensing, and Aerospace (CRSA 2022), Osaka, Japan, 26th August 2022
12. ***Keynote Speech***: "Name Signature Lookup System for Enhanced Security of NDN-based IoT Networks", International Faculty Development Program, Sharda University, Noida, India, 23rd June 2022
13. ***Keynote Speech***: "Activity Recognition Based Intelligent Sensory System for Smart Home Control", International Conference on 3D Immersion, Interaction and Multi-sensory Experiences (ICDIIME 2022), Madrid, Spain, 27th June 2022
14. ***Keynote Speech***: "Activity Recognition Based Smart Home Control System", International Conference on Algorithms, Microchips and Network Applications (AMNA 2022), Zhuhai, China, 18th February 2022
15. ***Keynote Speech***: "Recommendation-based approach for personalized control, Visual Comfort & Energy Efficiency in office buildings", International Conference on Computer Graphics, Artificial Intelligence and Data Processing (ICCAID 2021), Harbin, China, 24th December 2021
16. ***Keynote Speech***: "Named Data Networking Based Interoperable and Secure Framework for Heterogeneous IoT Networks", International Conference on Internet of Things and Machine Learning (IoTML 2021), Dalian, China, 17th December 2021
17. ***Keynote Speech***: "Security Enhancement to Named Data Networking", International Conference on Cyber Warfare, Security & Space Research (SpacSec'21), Jaipur, India, 10th December 2021
18. ***Invited Talk*****:** "Personalized Visual Comfort and Energy Efficiency in Office Buildings", International Conference on Sustainability and Equity: Digital Society, Bhubaneswar, India, 24th November 2020
19. ***Invited Talk*****:** "Personalized Visual Comfort and Energy Efficiency in Office Buildings", co-organized by IEEE Kharagpur Section and National Academy of Sciences India (NASI) Kharagpur Chapter, 16th July 2020

20. ***Keynote Speech***: "Named Data Networking: towards data-centric, fast and secure new generation Internet technology", 5th International Conference on Advanced Computing and Intelligent Engineering (ICACIE 2020), Mauritius, 25th June 2020
21. ***Keynote Speech***: "Named Data Networking: towards data-centric, fast and secure new generation Internet technology", 2nd International Conference on Information Technology and Electrical Engineering (ICITEE 2019), Changsha, Hunan, China, 6th December 2019
22. ***Invited talk*****:** "Underwater Sensor Network Protocols at Various Layers", Faculty Development Program, Haldia Institute of Technology, Haldia, India, 4th August 2016

## ❖ Conference Chair:

1. ***Publication Chair***: International Conference on Computer Technology and Biomedical Science (CTBS 2026), 14 - 16 August 2026, Beijing, China
2. ***Publication Chair***: International Conference on Machine Learning and Embodied Intelligence (ICMLEI 2026), 11–13 December 2026, Nanjing, China
3. ***Publication Chair***: 5th International Symposium on Sensor Technology and Control (ISSTC 2026), 16 – 18 October 2026, Qingdao, China
4. ***Track Chair***: 11th International Conference on Frontiers of Signal Processing (ICFSP 2026), 10-12 June 2026, Toulouse, France
5. ***General Chair:*** 4th International Conference on Advanced Communications and Machine Intelligence (MICA 2025), 22 – 24 December 2025, Tripura, India
6. ***General Chair***: Asia-Europe Conference on Cybersecurity, Internet of Things and Soft Computing (CITSC 2026), 10-12 January 2026, Rimini, Italy.
7. ***Track Chair***: 10th International Conference on Frontiers of Signal Processing (ICFSP 2025), 10-12 September 2025, Paris, France
8. ***General Chair***: Asia-Europe Conference on Cybersecurity, Internet of Things and Soft Computing (CITSC 2025), 10-12 January 2025, Rimini, Italy.
9. ***Publication Chair***: International Conference on Artificial Intelligence and Computer Information Science (AICIS 2024), 15-17 March 2024, Guangzhou, China.
10. ***Publication Chair***: 2nd International Conference on Computer, Internet of Things and Control Engineering (CITCE 2023), 3 - 5 November 2023, Leicester, UK.
11. ***Program Committee Chair***: International Conference on Images, Algorithms and Artificial Intelligence (ICIAAI 2023), 11 – 13 August 2023, Singapore
12. ***General Chair***: 2nd International Conference on 3D Immersion, Interaction and Multi-sensory Experiences (ICDIIME 2023), 27- 29 June 2023, Madrid, Spain.
13. ***Publication Chair***: 2nd International Conference on Computer, Internet of Things and Control Engineering (CITCE 2022), 11 - 13 November 2022, Sanya, China.
14. ***Publication Chair***: 2nd International Conference on Artificial Intelligence, Virtual Reality and Visualization (AIVRV 2022), 23 - 25 September 2022, Chongqing, China.
15. ***General Chair***: International Conference on Advanced Communications and Machine Intelligence (MICA 2022), 7-9 July 2022, Tamil Nadu, India.
16. ***General Chair***: International Conference on 3D Immersion, Interaction and Multi-sensory Experiences (ICDIIME 2022), 27-29 June 2022, Madrid, Spain.
17. ***International Advisory Chair*****:** 10th International Conference of Information and Communication Technology (ICICT2020), 24-25 April 2020, Wuhan, Hubei, China.

## ❖ Conference Session Chair:

1. 17th IEEE International Wireless Communications & Mobile Computing Conference (IWCMC 2021), June 2021, Harbin, China.
2. International Conference on Sustainability and Equity: Digital Society, 24 November 2020, Bhubaneswar, India,
3. 20th IEEE International Conference on Communication Technology (ICCT 2020), October 2020, Nanning, China.
4. ACM International Workshop on Future Industrial Communication Networks (FICN) in conjunction with ACM MobiCom, November 2018, New Delhi, India.
5. IEEE Region 10 Conference (TENCON 2016), November 2016, Singapore.

## ❖ Conference International Advisory Committee Member:

1. International Conference on Intelligent Systems and Industrial Instrumentation (CISII 2024), 20-21 September 2024, Kolkata, India
2. 3rd International Conference on Advanced Computing and Software Engineering (ICACSE 2021), 19-20 February 2021, Sultanpur, Uttar Pradesh, India.

3. International Conference on Research and Applications in Artificial Intelligence (RAAI 2020), 19-20 December 2020, Kolkata, India.
4. International Conference on Innovations in Computational Intelligence and Computer Vision (ICICV 2020), 17-19 January 2020, Jaipur, Rajasthan.
5. International Conference on Applied Mathematics and Computational Sciences (ICAMCS 2019), October 17-19, 2019, Dehradun, India.
6. International Conference on Advanced Computing and Software Engineering (ICACSE 2019), February 2019, Sultanpur, U.P., India.

## ❖ Conference Technical Program Committee Member:

1. IEEE International Smart Cities Conference (IEEE ISC2 2026), 27 – 30 October 2026, Porto, Portugal
2. 4th Workshop on Low-Cost Digital Solutions for Industrial Automation (LoDiSA 2026), 23 - 24 September 2026, University of Cambridge, UK.
3. IEEE International Conference on E-health Networking, Application & Services (IEEE Healthcom 2026), 19 - 21 October 2026, New York, USA.
4. IEEE International Conference on Communication: SAC E-Health Track (ICC2026:SAC-EH), 24–28 May 2026, Glasgow, Scotland, UK.
5. 25th IEEE International Conference on Bioinformatics and Bioengineering (BIBE 2025), 06-08 November 2025, Athens, Greece
6. IEEE Global Communications Conference (GLOBECOM 2025), 8 - 12 December 2025, Taipei, Taiwan
7. IEEE International Conference on E-health Networking, Application & Services (IEEE Healthcom 2025), 21–23 October 2025, Abu Dhabi, United Arab Emirates.
8. The IEEE International Conference on Internet of Things and Intelligence Systems (IoTaIS), 6-8 November, 2025, Bali, Indonesia
9. 3rd Workshop on Low-Cost Digital Solutions for Industrial Automation (LoDiSA 2025), 23 - 24 Sep 2025, University of Cambridge, UK.
10. The 6th International Conference on Artificial Intelligence, Robotics, and Control, 7 - 9 May 2025, Savannah, GA, USA
11. The IARIA Annual Congress on Frontiers in Science, Technology, Services, and Applications (IARIA Congress 2025), 6 - 10 July 2025, Venice, Italy
12. IEEE International Conference on Communication: SAC E-Health Track (ICC2025:SAC-EH), 8–12 June 2025, Montreal, Canada.
13. The 9th International Conference on Digital Signal Processing (ICDSP 2025), 21-23 February 2025, Chengdu, China
14. The 15th IEEE Global Information Infrastructure and Networking Symposium (IEEE GIIS2025), 25-27 February 2025, Dubai, UAE
15. IEEE International Conference on E-health Networking, Application & Services (IEEE Healthcom 2024), 18–20 November 2024, Nara, Japan.
16. The 10th International Conference on Cloud and Big Data Computing (CBDCom 2024), 5 - 8 November 2024, Malay, Philippines
17. The 14th International Conference on Software Technology and Engineering (ICSTE 2024), 16 - 18 August 2024, Macau, China.
18. IEEE Global Communications Conference: Communication & Information Systems Security (Globecom 2024 CISS), 8–12 December 2024, Cape Town, South Africa.
19. 2nd Workshop on Low-Cost Digital Solutions for Industrial Automation (LoDiSA 2024), 01 - 02 October 2024, University of Cambridge, UK.
20. The 9th International Conference on Information and Network Technologies (ICINT 2024), 24-26 May 2024, Tokyo, Japan.
21. The 7th World Conference on Computing and Communication Technologies (WCCCT 2024), 12-14 April 2024, Chengdu, China
22. IEEE International Conference on E-health Networking, Application & Services (IEEE Healthcom 2023), 15–17 December 2023, Chongqing, China.
23. IEEE International Conference on Communication: SAC E-Health Track (ICC2024:SAC-EH), 9–13 June 2024, Denver, CO, USA.
24. The 4th International Conference on Cloud and Big Data (CLBD 2023), 11 - 12 November 2023, Dubai, UAE.
25. The 15th International Conference on Networks & Communications (NeCoM 2023), 23 - 24 September 2023, Toronto, Canada
26. The 6th International Conference on Algorithms, Computing and Artificial Intelligence (ACAI 2023), 22 - 24 December 2023, Sanya, China
27. The 7th World Conference on Computing and Communication Technologies (WCCCT 2024), 22-24 March 2024, Chengdu, China.

28. The 12th IEEE International Conference on Communications, Networks, and Satellite (ComNetSat 2023), 23-25 November 2023, Malang, Indonesia
29. The 8th International Conference on Digital Signal Processing (ICDSP 2024), 23-25 February 2024, Hangzhou, China
30. International Conference on Communications, Computing and Artificial Intelligence(CCCAI 2023), 23-25 June 2023, Shanghai, China
31. IEEE International Conference on Communication: SAC E-Health Track (ICC 2022:SAC-EH), 28 May – 01 June 2022, Rome, Italy.
32. The 6th World Conference on Computing and Communication Technologies (WCCCT 2023), 6 – 8 January 2023, Chengdu, China
33. The 12th International Conference on Communication and Network Security (ICCNS 2022), 1- 3 December 2022, Beijing, China
34. IEEE International Conference on Cybernetics and Computational Intelligence (CyberneticsCom 2022), 16-18 June 2022, Malang, Indonesia.
35. IEEE Global Communication Conference: Communication & Information Systems Security (Globecom 2022:CISS), 4 – 8 December, Rio De Janeiro, Brazil.
36. The 23rd IEEE International Conference on High-Performance Switching and Routing (HPSR 2022), 6 - 8 June 2022, Taicang, Jiangsu, China.
37. The 14th IEEE International Conference on Internet of Things (iThings 2021), 06-08 December 2021, Melbourne, Australia.
38. IEEE Global Communication Conference: Communication & Information Systems Security (Globecom 2021:CISS), 7-11 December 2021, Madrid, Spain.
39. IEEE International Conference on Communication: SAC E-Health Track (ICC 2022:SAC-EH), 16 - 20 May 2022, Seoul, South Korea.
40. The 21st IEEE International Conference on Communication Technology (ICCT 2021), 13 - 16 October 2021, Tianjin, China.
41. The 10th IEEE International Conference on Communication, Networks, and Satellite (COMNETSAT 2020), 17 - 18 July 2021, Purwokerto, Indonesia.
42. The 6th EAI International Conference on Smart Grid and Innovative Frontiers in Telecommunications (Smart GIFT 2021), 12 - 13 August 2021, Virtual conference.
43. IEEE International Conference on Cloud and Big Data Computing (CBDCom 2021), 23 - 26 August 2021, Calgary, Canada
44. The 9th IEEE International Conference on Communication, Networks, and Satellite (COMNETSAT 2020), 17 - 18 December 2020, Batam, Indonesia.
45. International Conference on IoT and its Applications (ICIA 2020), December 2020, Jamshedpur, India
46. The 20th IEEE International Conference on Communication Technology (ICCT 2020), 28 - 31 October 2020, Nanning, China.
47. The 13th IEEE International Conference on Internet of Things (iThings 2020), 02 - 06 November 2020, Rhodes Island, Greece.
48. IEEE International Conference on Communication: SAC E-Health Track (ICC 2021: SAC-EH) and COVI-COM Workshop, 14 - 18 June 2021, Montreal, Canada.
49. IEEE Globecom 2020 Communication & Information Systems Security Symposium (Globecom 2020: CISS), 7 - 11 December 2020, Taipei, Taiwan
50. IEEE International Conference on Cloud and Big Data Computing (CBDCom 2020), 17 - 21 August 2020, Calgary, Canada
51. IEEE International Conference on Communication – Big Data Track (ICC 2020: BD Track), Dublin, Ireland, 7 - 11 June 2020.
52. The 11th IEEE International Symposium on UbiSafe Computing (UbiSafe 2019), Atlanta, USA, 14 - 17 July 2019.
53. The 6th International Symposium on Big Data Principles, Architectures, and Applications (BDAA 2019), Dublin, Ireland, 15 – 19 July 2019.
54. ACM International Workshop on Future Industrial Communication Networks (FICN) in conjunction with ACM MobiCom, New Delhi, India, November 2018
55. International Conference on Communication (ICC): SAC Symposium Smart Cities Track, Kansas City, MO, USA, 2018
56. IEEE International Smart Cities Conference (ISC 2017), Wuxi, China, 14 - 17 September 2017
57. The 86th IEEE Vehicular Technology Conference (VTC 2017), Toronto, Canada, 2017
58. International Conference on Intelligent, Secure, and Dependable Systems in Distributed and Cloud Environments (ISDD 2017), Canada, 2017
59. The 20th International Conference of Network-Based Information System (NbiS 2017), Canada, 2017.
60. International Conference on Advanced Computing and Intelligent Engineering (ICACIE 2016), Bhubaneswar, Odisha, India, 2016.

61. The 4th International Conference on Intelligent Human-Computer Interaction (IHCI 2012), IIT Kharagpur, Kharagpur, India, 2012.

## ❖ Short Term Course Organization:

1. ***Main organizer***: China Exchange Program for Young International (CEP) at the Faculty of Science and Technology, University of Nottingham Ningbo China, supported by the China International Talent Exchange Foundation (CITEF), 2 – 27 March 2026
2. ***Organizing team member*****:** International Summer and Winter Term (ISWT) course at the Department of Computer Science and Engineering, IIT Kharagpur, June 2015.
3. ***Organizing team member*****:** Short Term Course on Object-Oriented Programming and Data Structures using C++, Department of Computer Science and Engineering, IIT Kharagpur, 2 - 20 June 2014

## ❖ Administrative Duties:

1. Faculty Director of Academic Development and Outreach, Faculty of Science and Engineering, University of Nottingham Ningbo China, 2025 – Present
2. Faculty Director of Student Engagement and Feedback, Faculty of Science and Engineering, University of Nottingham Ningbo China, 2022 – Present
3. PGCHE faculty advisor of FoSE, Faculty of Science and Engineering, University of Nottingham Ningbo China, 2023 – Present.
4. Learning Community Forum (LCF) Coordinator, School of Computer Science, University of Nottingham Ningbo China, 2023 - Present
5. The BCS Accreditation Coordinator, School of Computer Science, University of Nottingham Ningbo China, 2023 – 2025
6. Computer Science UG Course Director, School of Computer Science, University of Nottingham Ningbo China, 2023 - 2025
7. Member of the Admission and International Relations Committee at the University of Nottingham Ningbo China, 2020 - 2021
8. Member of the EC Committee, School of Computer Science, University of Nottingham Ningbo China, 2020 - 2021
9. Held the post of Cashier at Vikram Sarabhai Residential Complex (VSRC) Hall, IIT Kharagpur, 2013 - 2014
10. Held the post of Block Representative at Vikram Sarabhai Residential Complex (VSRC) Hall, IIT Kharagpur**,** 2012 - 2013

## ❖ PhD Thesis Examination:

1. ***External Examiner*****:** Arup Sarkar, "Secure Healthcare System Using Blockchain", School of Computer Engineering, KIIT Deemed University, Odisha, India, 2025
2. ***External Examiner*****:** Syed Ali Hussain, "Design & Implementation of an Efficient Low Power Hybrid LIF Neuron Architecture with Spiking Neural Network for Neuromorphic Applications", School of Engineering and Sciences, SRM University - AP, Andhra Pradesh, India, 2025
3. ***External Examiner*****:** Sushil Kumar Mahapatra, "An IoE Based Approach to Improve Educational Outcomes in Virtual Classroom Environment", Department of Computer Science & Engineering, Siksha 'O' Anusandhan (Deemed to be University), Odisha, India, 2025
4. ***External Examiner*****:** Padmini Mansingh, "Health Care Data Management Using Big Data Analytics", Department of Computer Science & Engineering, Siksha 'O' Anusandhan (Deemed to be University), Odisha, India, 2024
5. ***Internal Examiner*****:** Yinglun LI, "Smart Health at Home: Multi-Sensor Monitoring and Disease Detection System for Cardiac Health Using Wearable Technology", School of Computer Science, University of Nottingham Ningbo China, 2025
6. ***Internal Examiner*****:** Lincan Li, "Mobile Edge Cache Technique Under Radio Access Network for the Next Generation of Communication Networks", Department of Electrical and Electronic Engineering, University of Nottingham Ningbo China, 2021

## ❖ Paper Review:

1. IEEE Transactions on Network and Service Management
2. IEEE Transactions on Vehicular Technology
3. IEEE Transactions on Parallel and Distributed Systems
4. IEEE Transactions on Mobile Computing
5. IEEE Transactions on Network Science and Engineering
6. IEEE Transactions on Sustainable Computing
7. IEEE Journal of Biomedical and Health Informatics
8. IEEE Systems Journal

9. IEEE Sensors Journal
10. IEEE Internet of Things Journal
11. IEEE Network
12. IEEE Communication Magazine
13. IEEE Internet of Things Magazine
14. Building and Environment (*Elsevier*)
15. Computer Networks (*Elsevier*)
16. Smart Health (*Elsevier*)
17. Results in Engineering (*Elsevier*)
18. IET Wireless Sensor Systems
19. Multimedia Tools and Applications *(Springer)*
20. Wireless Personal Communications *(Springer)*
21. Telecommunication Systems (*Springer*)
22. Journal of Supercomputing (*Springer*)
23. International Journal of Communication Systems (*Wiley*)
24. Security and Communication Networks (*Wiley*)
25. Journal of Ambient Intelligence and Smart Environments (IOS Press)
26. International Journal of Communication Networks and Distributed Systems (*InderScience*)
27. Photonic Network Communications
28. International Journal of Engineering Business Management
29. Journal of Mobile Multimedia (*River Publisher*)
30. IEEE Healthcom, 2023, 2024, 2025
31. IEEE Globecom, 2020, 2025
32. International Conference on Mining Intelligence and Knowledge Exploration (*MIKE*), 2013
33. IEEE Students' Technology Symposium (IEEE *TechSym*), 2014, 2016
34. IEEE International Conference on Communication (*ICC*), 2015, 2018, 2020, 2022, 2023, 2024, 2025
35. IEEE Vehicular Technology Conference (VTC), 2016
36. IEEE International Conference on Cyber-Physical and Social Computing (CPSCom), 2019

### ❖ Book Review

1. The seventh edition of "Database Systems: A Practical Approach to Design, Implementation, and Management" by Thomas Connolly and Carolyn Begg, Pearson Global Editions

## Students

### ❖ PhD Students:

#### ➢ Completed

1. Joshua Lai, "Multisensory Wearable Blood Pressure Monitoring System for Smart Health Home", School of Computer Science, University of Nottingham Ningbo China, 2021 – 2026 (*Lead Supervisor*)
2. Ningyuan Chen, "Scalable Blockchain Architectures for Enhancing Integrity and Privacy in Vehicular Ad-hoc Networks", Department of Electrical and Electronic Engineering, University of Nottingham Ningbo China, 2020 – 2024 (*Co-Supervisor*)

#### ➢ Pursuing

1. Tao Zhang, "Self-Evolving Edge Intelligence", School of Computer Science, University of Nottingham Ningbo China, 2026 – 2030 (*Lead Supervisor*)
2. Liang Deng, "Geospatially-Aware Semantic Routing for Space-Air-Ground Integrated Network", School of Computer Science, University of Nottingham Ningbo China, 2026 – 2030 (*Lead Supervisor*)
3. Ximing Zhang, "Exploring an optimal security mechanism for ubiquitous V2X connections", School of Computer Science, University of Nottingham Ningbo China, 2024 – 2028 (*Lead Supervisor*)
4. Wenzhuo Lyu, "Congestion Control in Named Data Networks", School of Computer Science, University of Nottingham Ningbo China, 2021 – 2027 (*Lead Supervisor*)
5. Yuancheng Zhang, "Human factors and collaboration in design", Department of Mechanical, Materials and Manufacturing Engineering, University of Nottingham Ningbo China, 2026 - 2030 (*Co-Supervisor*)
6. Chenyu Yang, "Resource Allocation in Track Platooning", Department of Electrical and Electronic Engineering, University of Nottingham Ningbo China, 2022 - 2026 (*Co-Supervisor*)
7. Qinglin Mao, "Text Image Inpainting for Recognition", School of Computer Science, University of Nottingham Ningbo China, 2022 – 2026 (*Co-Supervisor*)

### ❖ Master's Thesis Students:

#### ➢ Completed

1. Haozhe Geng, "Learner-based Teaching Plan by Considering Individual Mental Health Profile through Social Media Analysis", School of Computer Science, University of Nottingham Ningbo China, 2025 - 2026
2. Zhaokun Ding, "Behaviour-Adaptive PIT Management for Low-Latency NDN in Vehicular Networks", School of Computer Science, University of Nottingham, UK, 2024 - 2025
3. Cheng Yan, "Developing a data-secure postal system using Hyperledger Fabric smart contracts", School of Computer Science, University of Nottingham, UK, 2023 - 2024
4. Lihong Wei, "Optimization and Performance Evaluation of Dynamic Threshold-Based Handover Mechanism in Intelligent Transportation Systems", School of Computer Science, University of Nottingham, UK, 2023 - 2024
5. Wenchao Xia, "Lightweight Consensus Algorithms and the Layered Blockchain architecture applied in optimising for Resource-Constrained Devices", School of Computer Science, University of Nottingham, UK, 2023 – 2024
6. Zongzhe Li, "Blockchain-based secure and trustworthy Postal System", School of Computer Science, University of Nottingham, UK, 2022 - 2023
7. Ce Liu, "Blockchain-Based Secure NDN framework for Remote Health Monitoring System", School of Computer Science, University of Nottingham, UK, 2022 - 2023
8. Hao Li, "Optimization Strategies and Performance Evaluation of Health Monitoring Systems based on Named Data Networking (NDN) via Edge Intelligence", School of Computer Science, University of Nottingham, UK, 2022 - 2023
9. Shangqi Yang, "Designing an Innovative Intelligent Transportation Model Based on Digital Twin and Cyber-Physical Systems", School of Computer Science, University of Nottingham, UK, 2022 - 2023
10. Xuesong Zeng, "EIHR Management System Based on Blockchain and IPFS", School of Computer Science, University of Nottingham, UK, 2022 - 2023
11. Shuxin Feng, "Intelligent Traffic Prediction System by Combining Weather and Road Condition Information: a deep learning-based approach", School of Computer Science, University of Nottingham, UK, 2021 - 2022
12. Ayushman Sanyal, "Architecture for the deployment of C-V2X technology for different landscape requirements", School of Computer Science, University of Nottingham, UK, 2021 - 2022
13. Daiyaan Dharsey, "Development of a Blockchain-Based Electronic Patient Records (EPR) system with less vulnerable endpoints using asymmetric encryption and hashing", School of Computer Science, University of Nottingham, UK, 2021 - 2022
14. Harri Prasad, "Blockchain-Based Secure and Trustworthy Data Communication Model for Internet of Things", School of Computer Science, University of Nottingham, UK, 2020 - 2021

## ❖ Bachelor Thesis Students:

### ➢ Completed

1. Fangyuan Lu, "A Blockchain-based Consent Management System for Healthcare", School of Computer Science, University of Nottingham Ningbo China, 2025 – 2026
2. Ziyu Zhou, "Blockchain-Based AI Model Provenance", School of Computer Science, University of Nottingham Ningbo China, 2025 – 2026
3. Jiahao Liu, "A Blockchain and AI-Based System for Trustworthy Cross-Industry Document Verification", School of Computer Science, University of Nottingham Ningbo China, 2025 – 2026
4. Jiangbo Yu, "An Active DRL Data Center Network Content Delivery System Based on Lightweight Anomaly Detection", School of Computer Science, University of Nottingham Ningbo China, 2025 - 2026
5. Zhengkai Qiu, "A Genomic Data Analysis-Based Technique for Personalized and Precise Medicine", School of Computer Science, University of Nottingham Ningbo China, 2024-2025
6. Jiarui Wu, "TinyML-based Energy-Efficient Keyword Spotting on Resource-Constrained Devices Using MicroNets", School of Computer Science, University of Nottingham Ningbo China, 2024-2025
7. Feiyang Sun, "βFSCM: An Enhanced Food Supply Chain Management System Using Hybrid Blockchain", School of Computer Science, University of Nottingham Ningbo China, 2023-2024
8. Hangqi Wan, "Development of a Dynamic NDN Naming Scheme for Efficient and Fast FIB Lookup to Improve the Network Performance", School of Computer Science, University of Nottingham Ningbo China, 2023-2024
9. Maksim Kozlov, "Blockchain Consensus Protocol for Resource Constrained Internet of Things (IoT) Devices", School of Computer Science, University of Nottingham Ningbo China, 2023-2024
10. Puttipatt INGKASIT, "Development of an Efficient Control System for Smart Home", School of Computer Science, University of Nottingham Ningbo China, 2023-2024
11. Yuanli Zhu, "Living Cost Analysis Based on Accurate Salary Estimation Scheme by Using BigData Technique", School of Computer Science, University of Nottingham Ningbo China, 2023-2024

12. Haoran Cheng, “Development of an integrated architecture of Programmable Switches and Sketch Abstractions for Detection of Distributed Denial-of-Service”, School of Computer Science, University of Nottingham Ningbo China, 2022-2023
13. Zhuoran Bi, “Real-Time Blood Glucose Prediction and Hypoglycemia Detection for Diabetic Patients Using Multi-model Machine Learning Technique”, School of Computer Science, University of Nottingham Ningbo China, 2022-2023
14. Tianyi Ma, “Mental Health Assessment and Disease Prediction through Combination of Facial Expression and Voice”, School of Computer Science, University of Nottingham Ningbo China, 2022-2023
15. Yujie Chen, “A Blockchain-based Decentralised Web Hosting System using Peer-to-Peer File-Sharing Networks”, School of Computer Science, University of Nottingham Ningbo China, 2022-2023
16. Li-Kai WU, “Blockchain-based Fake Product Identification System”, School of Computer Science, University of Nottingham Ningbo China, 2021-2022
17. Longwen HU, “Design of Blockchain-based Secure Health Status Monitoring System Using Decentralized Machine Learning Technology”, School of Computer Science, University of Nottingham Ningbo China, 2021-2022
18. Dongchi Li, “B-Spot: Blockchain and Steganography-based Robust and Secure Photo Transmission Mechanism”, School of Computer Science, University of Nottingham Ningbo China, 2020-2021
19. Xiangning LIANG, “Robust and Secure File Transmission Through Video Streaming Using Steganography and Blockchain”, School of Computer Science, University of Nottingham Ningbo China, 2020-2021
20. Zhengrui Xue, “Identification of Fake Images on Social Media Platforms Using Big Data Analytics”, School of Computer Science, University of Nottingham Ningbo China, 2020-2021
21. Zhongyi Wang, “Identifying Covid-19 Fake News Using Deep Learning”, School of Computer Science, University of Nottingham Ningbo China, 2020-2021
22. Zhenbang YAN, “GUI-based Network Simulator”, School of Computer Science, University of Nottingham Ningbo China, 2020-2021
23. Zhao Peng, “Activity-based home appliance control system”, School of Computer Science, University of Nottingham Ningbo China, 2019-2020

## ❖ Group Project Students

### ➢ Completed

1. Yao Xiao, Chenyuan Liu, Si Yu Chang, Zeng Cai, Yi-Ting Peng, and Yuyi Ye, "LLM-based University Finder System", School of Computer Science, University of Nottingham Ningbo China, 2025 - 2026
2. Chengyu Zhao, Jin Fu, Julienne Adwin, Kaifeng Qin, Yucheng Peng, Zhiwei The, “Body Language Detection-Based Emotionally Controlled Music Player”, School of Computer Science, University of Nottingham Ningbo China, 2024 - 2025
3. Bokai Zhang, Xinyi Song, Su Yang, Francesco Tedjo, Xinran Jiang, “Accurate Student Evaluation of Teaching System”, School of Computer Science, University of Nottingham Ningbo China, 2023 - 2024
4. Chongpo Shao, Jianing Wu, Yuanli Zhu, Feiyang Sun, Hongman Li, Yaru He, “Find a Doctor”, School of Computer Science, University of Nottingham Ningbo China, 2022 - 2023
5. Yichi Zhang, Chengyue Pan, Baizhen Lin, Kunning Wang, and Tianshu Ye, “Feedback Analysis Based Product Rating System”, School of Computer Science, University of Nottingham Ningbo China, 2021 – 2022
6. Haowen Zheng, Honghao Zeng, Tianqi Xia, Boyuan Ye, Yuhua Guo, Zeyu Wang, “Epidemic disease control using the Internet of Things and Big Data analytics”, School of Computer Science, University of Nottingham Ningbo China, 2020 - 2021.
7. Xiaoman Ding, Xiaoning Ma, Yunzhe Dong, Minxiao Li, Yixuan Wang, Yixuan Li, “Plug-n-Play Secure Remote Health Monitoring System”, School of Computer Science, University of Nottingham Ningbo China, 2019 - 2020.

## ❖ Internship Project Students

### ➢ Completed

1. Yijun Sun and Hanbo Zhao, "Edge Computing in Content Centric Network (CCN)", School of Computer Science, University of Nottingham Ningbo China, June – August 2026
2. Yijun Sun, Yuxuan Si, Yue Wu, and Qiaoyi Zhu, "Interoperability and Softwarization of Content Centric Networking (CCN) for Internet of Things", School of Computer Science, University of Nottingham Ningbo China, March – June, 2026
3. Yan Ge, Zefei Xie, and Junxi Wu, "Artificial Intelligence in Content Centric Networking", School of Computer Science, University of Nottingham Ningbo China, June - August 2024

4. Shuhang Kong and Huayi Zhang, "Blockchain-based Content-Centric Networking", School of Computer Science, University of Nottingham Ningbo China, June - August 2024
5. Run Yang and Hangqi Wan, "Mobility and Security issues of Content-Centric Network and their solution", School of Computer Science, University of Nottingham Ningbo China, June - August 2023
6. Zhihang Zhu and Yiran Mang, "Recommender System for Healthcare", School of Computer Science, University of Nottingham Ningbo China, March - June 2022
7. Tianyi Ma and Chenyu Yang, "Research and Development of Trust-Centric and Attack-Centric Recommender Systems", School of Computer Science, University of Nottingham Ningbo China, March - June 2022
8. Xinyi Wang and Yifei Miao, "Recommender System Applications", School of Computer Science, University of Nottingham Ningbo China, March - June 2022
9. Lin Chen, Weixue Sheng, "An efficient technique for identification of cache pollution attack in NDN-based remote health monitoring system using rank comparison", School of Computer Science, University of Nottingham Ningbo China, February-August 2021
10. Jiayi Shi, Kewei Chen, "DMACN - A Dynamic Multi-Attribute Caching Mechanism for NDN-Based Remote Health Monitoring System", School of Computer Science, University of Nottingham Ningbo China, February-August 2021.
11. Rubin Chen, Yukai Qian, "Efficient Technique for Real-time Communication Between Doctor and Patients in an NDN-based Remote Health Monitoring System", School of Computer Science, University of Nottingham Ningbo China, February-August 2021.
12. Hanyu Su, Siqi Yang, Yuning SUN, Hongru ZHAO, "Estimation of salary increment of employees in an organization by living cost analysis using BigData", School of Computer Science, University of Nottingham Ningbo China, June-August 2021
13. Leyang Hu, Yichen Huang, "An Integrated Framework for a Single Device to Combine Face Recognition and Covid-19 Detection Using Thermal Infrared Imagery", School of Computer Science, University of Nottingham Ningbo China, June - August 2021
14. Yihan Zhang, Xiyu Du, Peiyu Wang, "*βFSCM*: An Enhanced Food Supply Chain Management System Using Hybrid Blockchain", School of Computer Science, University of Nottingham Ningbo China, June – August 2021
15. Ruibin Chen, Yachen Zhang, Yichen Fei, "*WLEACH-CK*: Weighted K-Means Based LEACH-C Algorithm for Cluster Head Selection", School of Computer Science, University of Nottingham Ningbo China, June – August, 2020
16. Jiayi Shi, Weixue Sheng, "A Novel Framework for Predicting the Spread of COVID-19 by Contact Tracing through Smartphone", School of Computer Science, University of Nottingham Ningbo China, June – August, 2020.
17. Zhicheng Song, "Name-Signature Lookup System: A Security Enhancement to Named Data Networking", School of Computer Science, University of Nottingham, UK, June – August, 2020
18. Shanglun Wu, Yujie Yuan, "Lightweight Verification and Fine-grained Access Control in Named Data Networking Based on Schnorr Signature and Hash Functions", School of Computer Science, University of Nottingham Ningbo China, June – August, 2020.
19. Donglin Jiang, Zhihui Zhang, "Efficient Cache Replacement for High Availability of Data Content from Content Store in An NDN Node Based on LSTM and Knowledge Distillation", School of Computer Science, University of Nottingham Ningbo China, June – August, 2020.
20. Yunuo Zhao, Hao Jiang, "Optimizing Open Shortest Path First Routing Calculation for NDN Using Johnson Algorithm", School of Computer Science, University of Nottingham Ningbo China, June – August, 2020.

## Collaborations

1. Prof. Hao Wang, Professor, Xidian University, China
2. Dr. Ayan Mondal, Assistant Professor, Department of Computer Science and Engineering, Indian Institute of Technology Indore, India
3. Dr. Arun Kumar, Assistant Professor, Department of Computer Science and Engineering, National Institute of Technology Rourkela, Odisha, India.
4. Dr. G. G. Md. Nawaz Ali, Assistant Professor, Department of Computer Science and Information Systems, Bradley University, IL, USA.
5. Dr. Monideepa Roy, Associate Professor, School of Computer Engineering, Kalinga Institute of Industrial Technology, Odisha, India
6. Mr. Hans Ni, Project Leader, Deqing Ispatial Ltd. co, China

## Countries Visited

England, Singapore, Malaysia, Indonesia, Philippines, Norway, Czech Republic, Austria, Switzerland, Italy, Netherlands, Belgium, France, Spain, China

## Personal Details

**Date of Birth:** 14th March 1983.
**Sex:** Male.
**Marital Status:** Married
**Nationality:** Indian.
**Languages Known:** Bengali (Mother tongue), English (Proficient) & Hindi (Proficient).
**Hobbies:** Reading storybooks, watching films.

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